

lesson plan: Synsets (defs) -> Lemmas (words) are part of those -> what other relationships do we have? -> semantic similarity

- ~10 mins until first group exercise (slow down...)
- ~5 mins for exercise 1
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I thought it was better to start from the concept of a synset because that's what WordNet is centered around. That might be a little confusing, so I should pause and make sure that made sense to everyone.

Ask for class input with bank's definitions: - place to deposit a check - riverbank - piggy bank - bank of switches

antonymy - doesn't always make sense on a conceptual level I would guess that it stems from the fact that lemmas are defined at a more specific level within synsets. Technically, hot is represented as many different lemmas, depending on which synset/definition you're looking at (notice how our jump and leap lemmas from earlier are both within the jump.n.01 synset). Based on how we actually use words in English, not everything within hot (meaning high temperature) has an opposite - 'hot' & 'cold' are antonyms, but what about 'scalding'?

What are all the relations that WordNet keeps track of? - synonyms (synsets) - antonyms - hyper/hyponyms (specificity of a concept) - entailment (requirement; X requires Y) - meronym/holonym (part/whole) - pertainym - troponym - more?

how exactly does WordNet calculate similarity? $1 / (\text{dist} + 1)$ - very few values close to 1; similarity scores fall off quickly - can we rescale that somehow? $2k / (i + j + 2k)$ - large k means there's a bias towards 1 - small k means there's a bias towards 0 / $(i + j)$

semantic similarity group exercise - embeddings; categories were broad enough that you'll probably have lots of unrelated words in related headline titles - WordNet; it's centered around synonym sets, so it can suggest other things within the same synset - WordNet; with a medical encyclopedia, I think being understand how it got to the results it did is important - embeddings; too broad for WordNet

creative synthesis ideas